Developer Handoff — Madagascar Freshwater Fish Registry

Audience: the engineering team implementing the redesigned registry in the production codebase. **Scope:** what the prototype shows, which parts are placeholder, and what must be wired to live data / real assets before shipping.

The prototype (`Madagascar Freshwater Fish redesign.html`) is a self-contained React + Babel app with dummy SVG silhouettes and a hand-drawn SVG island. **None of those visual assets ship.** This document tells the coding team which real data sources, tiles, and image libraries to point the page at instead.

---

1. Components that need real assets

Prototype component	File	What it renders now	What to wire it to
`MadagascarOutline`	`prototype/primitives.jsx`	Hand-drawn SVG `<path>` approximating the island, with scattered `<circle>` locality dots	**Remove the component.** Replace with the existing site map (see §2).
`FamilySilhouette`	`prototype/primitives.jsx`	One generic silhouette per fish family (`Cichlidae`, `Bedotiidae`, `Aplocheilidae`, `Gobiidae`, `Eleotridae`, `Anchariidae`) hard-coded as inline SVG	Replace with genus- or species-level silhouette/illustration pulled from the CMS/media library. See §3.
`LogoMark`	`prototype/primitives.jsx`	Placeholder mark in the nav	Swap for the official WPS registry mark.

Everything else (IUCN pills, tables, filters, profile layout, dashboard cards) is structural and works against the real data schema as-is.

---

2. Map: replace `MadagascarOutline` with the existing site map

The prototype draws a cartoon island and fakes dots. Production should use the same interactive map already embedded elsewhere on wildlifeprotectionsolutions.org.

Where the outline is used

Search/replace these call sites in the port:

- `prototype/pages-home.jsx:115` — small map inside the hero's Red-List-breakdown block
- `prototype/pages-home.jsx:226` (`HeroLiveMap`) — large map in the "live map" hero variant
- `prototype/pages-home.jsx:369` — tiny map inside the home silhouette grid

- `prototype/pages-other.jsx:209` — large map on the `/map` page
- `prototype/pages-profile.jsx:136` — species-profile "where it lives" map

Each instance passes a `dots` prop (array of `{id, x, y, iucn}`) and sometimes `highlightId`. Replace the whole component with the same map widget used on the live site. Contract we need from it:

```
<RegistryMap
  height={number}                  // container height in px
  species={Species | null}         // optional — when set, filter to that species'
localities
  highlightLocalityId={string?}    // for hover from the species list
  onLocalityClick={(loc) => void} // navigate to /species/:id
  visibleIucnCategories={Set<IucnCode>}  // legend toggles, map page only
/>
```

Data source for locality points must be the **canonical locality table** from the registry backend — not generated from basin centroids like the prototype does. If the site map currently reads from a GeoJSON endpoint (e.g. `/api/localities.geojson`), expose a filter param for species ID and IUCN category and use that.

Size variants to support

- **Hero thumbnail:** ~160×320
- **Profile panel:** ~180×360
- **Full map page:** fills container, ~360×720 or larger
- **Silhouette-grid thumbnail:** ~64×124 (dots only, no interaction)

If the existing map widget can't easily render at thumbnail sizes, use a static PNG/SVG of the site's basemap for those and reserve the interactive widget for the map page + profile.

Legend + dot styling

Use IUCN category colors from `prototype/primitives.jsx` → `window.IUCN_COLORS` (already aligned with the official IUCN palette). These must match the colored pills on species cards so the whole system reads as one chart.

---

3. Fish silhouettes: pull from the registry media library

Current behavior

`FamilySilhouette({ family })` returns one of six inline SVGs keyed on `family`. Every cichlid renders the same silhouette. This is wrong for a taxonomy product.

Required behavior

Create a `<SpeciesSilhouette />` component that resolves in this order:

1. **Species-level silhouette** — preferred. `species.silhouette_url` from the DB.
2. **Genus-level fallback** — e.g. all `Paretroplus` share one silhouette if species art is missing.
3. **Family-level fallback** — final backstop. Matches what the prototype currently does.

Schema addition on the `species` table / CMS model:

```
silhouette_url            text   -- /media/silhouettes/{genus}-{species}.svg
silhouette_credit         text   -- optional attribution
hero_photo_url            text   -- optional full-color photo for profile header
hero_photo_credit         text
```

And on the `genera` + `families` tables:

```
silhouette_url            text
```

Component contract

```
<SpeciesSilhouette
  species={Species}          // we read species.genus, species.family as fallback
keys
  variant="line"|"fill"      // line-art on cards, filled silhouette on hero
  height={number}            // width auto from aspect ratio
/>
```

Render as an `<img>` (or inline `<svg>` if the asset is SVG and we need theming). Always include `alt={species.scientific_name}`.

Call sites to update

Every `<FamilySilhouette family={sp.family} />` in the prototype becomes `<SpeciesSilhouette species={sp} variant="line" height={...} />`:

- `prototype/pages-directory.jsx:295` — species card thumb (roomy card)
- `prototype/pages-home.jsx:181, 211, 319, 357` — home page silhouette grid + hero
- `prototype/pages-other.jsx:257` — map page species list
- `prototype/pages-profile.jsx:64` — profile header scale indicator
- `prototype/chrome.jsx:87` — global search result row

Asset pipeline

- Silhouettes live under `/media/silhouettes/` (or equivalent in the CMS).
- SVG preferred, monochrome, transparent background, stroke weight consistent across the set so the grid reads as a system. Target ~200px wide at 1x.
- Sourcing plan (confirm with curators):
  - Flagship ~20 species get custom illustration.
  - Remaining ~~130 species fall back to genus silhouettes (~~25 genera).
  - Every family must have a final fallback (6 families currently: Cichlidae, Bedotiidae, Aplocheilidae, Gobiidae, Eleotridae, Anchariidae, plus any added).
- Credit line stored per asset; surface it in the profile page and in a site-wide attributions page.

---

4. Data model the prototype assumes

The prototype's `prototype/data.js` is seeded with ~20 representative records. The production schema should cover every field referenced. Fields the UI reads:

```
Species {
  id                    int (primary key, stable for URLs)
  scientific_name       string                    -- "Paretroplus menarambo"
  provisional_name      string?                   -- for undescribed morphospecies
  common_names          string[]?                 -- currently missing; see §5
  family                string
  genus                 string
  authority             string?                   -- "Allgayer"
  year_described        int?
  taxonomic_status      "described" | "undescribed_morphospecies"
  endemic_status        "endemic" | "native" | "introduced"
  iucn_status           "CR"|"EN"|"VU"|"NT"|"LC"|"DD"|"NE" | null
  iucn_criteria         string?                   -- "B1ab(iii,v)"
  cares_status          "CCR" | "priority" | "monitored" | "" -- CARES Conservation
Priority tiers
  shoal_priority        boolean                   -- SHOAL "priority species" flag
  max_length_cm         number?
  habitat               string                    -- free text, one line
  basin                 string                    -- canonical watershed name; join to
basins table
  description           string                    -- short natural-language summary
  has_husbandry         boolean                   -- any captive breeding docs on file
  has_localities        boolean                   -- any geo records on file
  localities            int                       -- count of records
  ex_situ: {
    institutions        int                       -- # holding institutions
    individuals         int                       -- total individuals in program
    programs            int                       -- # coordinated breeding programs
  }
  silhouette_url        string?                   -- §3
  hero_photo_url        string?                   -- §3
}
```

Aggregate/derived values used by the dashboard and home hero:

- Red-list histogram: `count(species) group by iucn_status`
- **Coverage gap** (shown in stat hero): `count(species where iucn_status in ('CR','EN','VU') and ex_situ.institutions = 0)`, divided by `count(threatened species)`
- Family breakdown for silhouette grid

Expose as one JSON endpoint the registry page hydrates from, e.g. `GET /api/registry/species.json` returning the full list + precomputed aggregates.

---

5. Fields to restore on the Species Profile

The prototype currently hides a few fields we said we want back. Make sure the profile shows them:

- **Common names** (`species.common_names[]`) — currently not in the seed data; needs DB column + editor UI.
- **Husbandry notes** — link out to or inline a rich-text `husbandry_notes` field. Show only when `has_husbandry` is true.

- **Field programs** — join to a `field_programs` table: `{ id, species_id, partner, region, started_at, url }`. Display as a small list.
- **Full ex-situ roster** — when the user expands, show the per-institution table, not just the roll-up numbers. Join to `ex_situ_holdings { species_id, institution, individuals, started_at, studbook_url? }`.

Layout for these sections is already in `prototype/pages-profile.jsx`; the data bindings just need to replace placeholders.

---

6. Dashboard: reorient to ex-situ coordinator audience

The dashboard currently shows Red List counts. Restructure around what an ex-situ coordinator needs to triage work:

1. **Coverage gap panel** — threatened species with no captive population, sortable by IUCN category and by basin.
2. **Single-institution risk** — species where `ex_situ.institutions ≤ 2`. These are one-aquarium-fire from extinction.
3. **Studbook status** — species with a coordinated breeding program vs. ad-hoc holdings.
4. **Recently added / updated** — last N records touched, for curator QA.
5. **Red List snapshot** — demoted from hero; lives at the bottom as context.

Data for all of this is derivable from the Species + ex-situ tables in §4.

---

7. Visual treatment for CARES / SHOAL flags

The prototype shows CARES tier and SHOAL priority as colored badges. Soften these: they should be a quiet text label or a small outlined pill next to the IUCN pill — not competing with it for attention. IUCN status is the primary signal; CARES and SHOAL are secondary programmatic flags.

CSS-wise: use `border: 1px solid var(--rule-strong)` + `color: var(--ink-2)`, no background fill. Same type scale as the metadata eyebrow.

---

8. Theming / tweak surface

The prototype exposes a Tweaks panel (palette, density, type system, IUCN color treatment). **Do not ship this** — pick the default theme (Journal colors + Field Guide type + muted-pastel IUCN + roomy density) and bake it into production CSS tokens. The other combinations were exploration only.

Tokens to promote to the production stylesheet live in `prototype/themes.js` and `prototype/styles.css`.

---

9. Routing

Prototype is hash-routed for embedding. Production should use real paths:

Prototype	Production
`#/home`	`/registry`
`#/species`	`/registry/species`

`#/species/:id`	`/registry/species/{slug-or-id}`
`#/map`	`/registry/map`
`#/map?species=:id`	`/registry/map?species={id}`
`#/dashboard`	`/registry/dashboard`
`#/about`	`/registry/about`

Use species `scientific_name` slugified (`paretroplus-menarambo`) as the canonical URL key and keep numeric ID as an alias.

---

10. Accessibility checklist

- Every silhouette `<img>` needs `alt={species.scientific_name}` .
- Map dots need keyboard focus + screen-reader labels ("Paretroplus menarambo, Bemarivo basin, Critically Endangered").
- IUCN pills must not rely on color alone — the two-letter code is already in the pill, keep it visible.
- CARES / SHOAL softened pills: include `aria-label="CARES Conservation Research"` etc.
- Min 44px hit targets on the filter chips and map legend toggles.

---

11. What's in the prototype that is intentionally throwaway

- The hand-drawn island SVG and all basin centroid math in `localitiesForSpecies` / `allLocalityDots` .
- The six placeholder family silhouettes inside `FAMILY_SILHOUETTES` in `primitives.jsx` .
- The Tweaks panel and all of `prototype/themes.js` 's alternate palettes.
- The `LogoMark` placeholder.
- Lorem-ipsum-ish descriptions on undescribed morphospecies — real copy from curators required.

Everything else (layouts, filter logic, card hierarchy, profile sectioning, dashboard groupings, copy tone) is intended to carry over.

Part 2 — Visual System

Everything above is data + plumbing. This section pins down the look: tokens, type, components, and page compositions. Pair this with the reference screenshots under `handoff-refs/` .

12. Design tokens (production CSS)

Promote these to your production CSS as custom properties on `:root` . The prototype explored three themes; the one below is the locked-in default ("Scientific Journal" palette + "Field Guide" type + muted pastel IUCN).

```
:root {
  /* — Surfaces — */
  --bg:          #FAFAF7;   /* page */
  --bg-raised:  #FFFFFF;   /* cards, inputs, dropdowns */
```

```
  --bg-sunken:  #F0EEE6;   /* section bands, footer */

  /* — Ink — */
  --ink:        #0F1412;   /* primary text, headings */
  --ink-2:      #2E3834;   /* body, metadata */
  --ink-3:      #6B7670;   /* captions, eyebrows, placeholder */

  /* — Rules / borders — */
  --rule:         #DCD9CE; /* card borders, dividers */
  --rule-strong:  #A8A598; /* input borders, buttons */

  /* — Accent (conservation green) — */
  --accent:       #2B6E5F; /* primary actions, links */
  --accent-2:     #0F2A2E; /* hover */
  --accent-soft:  #D6E6E0; /* nav active pill background */

  /* — Supporting — */
  --terracotta:   #A84420; /* logo Madagascar fill */
  --highlight:    #D4A84B; /* lichen gold — rare call-outs */
  --focus:        #2B6E5F;

  /* — Type families — */
  --serif: 'Spectral', 'Charter', Georgia, serif;
  --sans:  'IBM Plex Sans', system-ui, sans-serif;
  --mono:  'IBM Plex Mono', ui-monospace, monospace;

  /* — Shape — */
  --radius:     2px;       /* buttons, pills, inputs */
  --radius-lg:  3px;       /* cards, panels */

  /* — Shadows — */
  --shadow-sm: 0 0 0 1px rgba(15,20,18,0.04);
  --shadow:    0 1px 2px rgba(15,20,18,0.04), 0 0 0 1px rgba(15,20,18,0.06);

  /* — Heading overrides — */
  --h1-weight:   500;
  --h1-tracking: -0.015em;

  /* — Eyebrow treatment — */
  --eyebrow-case:     uppercase;
  --eyebrow-tracking: 0.22em;
}
```

Small-caps eyebrow (nice touch, used site-wide):

```
.eyebrow {
  font-family: var(--sans);
  font-size: 11px;
  font-weight: 600;
  color: var(--ink-3);
  font-variant-caps: all-small-caps;  /* only if the chosen sans supports it */
```

```
  font-feature-settings: "smcp";
  text-transform: none;
  letter-spacing: 0.12em;
}
```

---

13. Typography spec

We use three families with strict roles:

Role	Family	Weight	Size	Tracking	Notes
Display H1	Spectral	500	44–48px	-0.015em	Page hero heading
Section H2	Spectral	500	28px	-0.015em	Section openers
Subsection H3	Spectral	500	20px	-0.015em	Card / panel titles
H4	Spectral	500	16px	normal	Rare
Body	IBM Plex Sans	400	15px / 1.55	normal	Primary reading text
Body large	IBM Plex Sans	400	18px / 1.55	normal	Hero standfirsts
Text small	IBM Plex Sans	400	13px / 1.5	normal	Metadata rows, buttons
Text xs	IBM Plex Sans	500	11px / 1.4	normal	Pills, chip labels
Eyebrow	IBM Plex Sans small-caps	600	11px	0.12em	Section kickers
Caption	IBM Plex Sans	400	12px / 1.5	normal	Image + chart captions
Scientific name	Spectral italic	400–500	matches context	normal	Always italic; genus capitalized, species lowercase
Numerals (stats)	Spectral	500	48–180px	-0.02 to -0.04em	`font-variant-numeric: tabular-nums`
Mono (codes, coords)	IBM Plex Mono	400	11–13px	normal	IUCN codes in running text, lat/long

Rules:

- **Headings are always the serif.** No bold; 500 weight reads as "medium" on Spectral.
- **Body is always the sans.**
- **Italic is reserved for scientific names.** Never italicize body copy.

- **Don't mix weights inside a line** (no bold emphasis on top of sans 400 except for a single `<strong>` per sentence at most).
- Numerals in stats and tables always tabular.

---

14. Color token usage rules

Token	Use	Don't
`--accent` (#2B6E5F)	Primary CTAs, links, active nav, focus ring	Huge backgrounds; hover states (use `--accent-2`)
`--accent-soft`	Nav active pill, tag highlights	Card backgrounds
`--ink`, `--ink-2`, `--ink-3`	Strict 3-step hierarchy for all text	Inventing a 4th gray
`--rule`, `--rule-strong`	1px borders (rule = cards, rule-strong = inputs + buttons)	Divider thickness >1px
`--terracotta`	Only the Madagascar silhouette in the logo mark	Anywhere else — it's brand only
`--highlight` (gold)	Rare "something to notice" accent (e.g. undescribed-morphospecies tag)	As a primary UI color
IUCN colors	Only on IUCN pills, Red-List-related charts, and map dots	Buttons, borders, or generic decoration

Never introduce new colors outside this set. If something feels like it needs color, it's probably an IUCN category or a CARES flag — use the official palette below.

IUCN Red List palette (official)

```
--iucn-ex:  #000000; /* Extinct */
--iucn-ew:  #542344; /* Extinct in the Wild */
--iucn-cr:  #D81E05; /* Critically Endangered */
--iucn-en:  #FC7F3F; /* Endangered */
--iucn-vu:  #F9E814; /* Vulnerable (text goes dark on this) */
--iucn-nt:  #CCE226; /* Near Threatened */
--iucn-lc:  #60C659; /* Least Concern */
--iucn-dd:  #D1D1C6; /* Data Deficient */
--iucn-ne:  #FFFFFF; /* Not Evaluated (rendered as outlined) */
```

Each category also has a `soft` background and `line` border for pill use — see §15.

---

15. Component spec

15.1 IUCN pill

```
[●] CR          ← small size, on cards and inline
```

- Height 20px small / 26px large
- Inline-flex, gap 6px, font 11px bold, letter-spacing 0.05em
- Border-radius 999px
- Left dot 8px circle
- **Solid fill** for CR and EN (white text on `--iucn-cr` / `--iucn-en`)
- **Soft fill** for VU, NT, LC, DD (soft background, colored text, 1px colored border)
- **NE** is outlined only (transparent fill, `--rule` border)
- Always includes the two-letter code; full label only when `showLabel` prop is true (profile header)

15.2 Species card (roomy — default)

Structure left → right inside a `.card`:

1. **3px full-height color bar** on the left edge, colored by IUCN category
2. **Silhouette column** (72px wide) — `<SpeciesSilhouette variant="line" />`, vertically centered
3. **Main column** (flex 1):
   - Scientific name, Spectral italic, 19px
   - Primary common name (if any), 13px, ink-3
   - Metadata row, 11px ink-3: `Family · Endemic-status · Basin`
4. **Right column**: IUCN pill top-aligned, plus CARES/SHOAL soft labels below

Card chrome:

- Padding: `18px 20px` (roomy), `14px 16px` (default), `10px 12px` (compact)
- Border: 1px `--rule`, radius `--radius-lg` (3px)
- Hover: border shifts to `--ink-2`, subtle shadow `--shadow`
- Whole card is a `<button>` — entire surface clickable to the profile

Grid:

- Roomy → single column, 12px gap
- Default/compact → 2 columns, 14px gap

15.3 Filter chips

- `IBM Plex Sans` 11px, weight 500 inactive / 600 active
- Pill: `padding: 5px 10px`, `border-radius: 999px`
- Inactive: `--bg-raised` fill, `--rule-strong` border, `--ink-2` text
- Active: `--accent-soft` fill, `--accent` border, `--accent-2` text
- Gap 4px, wrap freely; group by filter type with 11px eyebrow label above each group

15.4 Segmented control (Red List category, endemic status)

Same visual as filter chips but grouped tightly in a single row with no gap between siblings — use when options are mutually exclusive.

15.5 Inputs

- 13px IBM Plex Sans, 32px total height, 1px `--rule-strong` border, `--radius` (2px)
- Focus: border `--accent`, 3px ring at 20% opacity accent
- Search input: 32px left padding to clear the magnifier icon (`--ink-3`, 14px)

15.6 Buttons

Variant	Background	Border	Text	Hover
`btn` (default)	`--bg-raised`	`--rule-strong`	`--ink`	border → `--ink-2`, bg → `--bg`
`btn-primary`	`--accent`	`--accent`	white	`--accent-2` on both
`btn-ghost`	transparent	transparent	`--ink`	bg → `--bg-sunken`

All buttons: `IBM Plex Sans` 13px weight 500, padding `7px 12px`, radius `--radius` (2px).

15.7 Basin pill (used on profile header + map hover)

- 11px uppercase, letter-spacing 0.06em
- `padding: 2px 8px`, 1px `--rule` border, `--bg` fill, `--ink-3` text
- Radius 999px

15.8 Coverage-gap stat treatment (hero headline numeral)

- Big numeral: Spectral 500, 180px, `letter-spacing: -0.04em`, `line-height: 0.9`, tabular
- "of 55" pair to the right, baseline-aligned: `of` at 13px in `--ink-3`, the 55 at 48px Spectral 500 in `--ink-2`
- Below: standfirst at 18px `--ink-2`, max-width 460px
- Eyebrow above in IUCN CR red: "Coverage gap"

---

16. Page compositions

16.1 Site header (sticky)

- Height 56px, sticky with `backdrop-filter: saturate(120%) blur(8px)`
- Background: `color-mix(in oklab, var(--bg) 90%, transparent)`
- Bottom border 1px `--rule`
- Layout: `[Logo] [Nav items] [flex] [Search 280px] [Tweaks button]`
- Max width 1280px, horizontal padding 28px, gap 20px
- Nav item active: `--accent-soft` background pill with `--accent` text, weight 600

16.2 Home / Welcome page

A single page, four sections top → bottom. **Use the "stat" hero (coverage gap) as the default** — it's the one the organization tested best against.

Section A — Hero (stat variant)

- Container padding `56px 28px 64px`
- Small eyebrow line: "Madagascar Freshwater Fish · Conservation Registry"
- H1 (Spectral 48px / 1.08 line-height / 500, max-width 680px): *"A shared record for Madagascar's endemic freshwater fish."*
- 2-column grid 56px gap, `1.1fr 0.9fr`:
  - **Left:** coverage-gap stat (§15.8) + standfirst + primary & secondary CTAs
  - **Right:** map thumbnail (160×320, from the real WPS map widget per §2) + Red List breakdown histogram (see §16.2a)
- Bottom border 1px `--rule`

Section A histogram (right side of hero)

- Eyebrow "Red List breakdown" above
- Stacked rows, each: `[two-letter pill][thin horizontal bar][count]`
  - Bar length proportional to count / total
  - Bar color = IUCN category color
  - Bar height 10px, background `--rule` behind for missing width
- 6px row gap

Section B — "A shared record"

- Container padding `56px 28px`
- Eyebrow "A shared record", H2 (28px Spectral, max-width 760px), short paragraph (15px, max-width 680px)
- Three-up card grid below, gap 20px:
  - 01 — Species directory
  - 02 — Distribution map
  - 03 — Conservation dashboard
- Each card: eyebrow number, H3, one-line body, visual preview (140px tall), CTA link at bottom

Section C — Last-sync strip

- 20px 28px padding, `--bg-sunken` background, top border `--rule`
- Row: `[pulsing dot + "Live"] [sync log sentence] [flex] [See sync log →]`
- Pulsing dot: 8px accent circle with a 4px rgba outer halo

16.3 Species Directory

- Two-column layout: `240px` filter rail, flexible content area, 32px gap
- Container padding `40px 28px`
- Top bar: H1 "Species directory" + result count + view-density segmented control ("Compact / Default / Roomy") + sort dropdown
- Left rail filter groups (each with 11px eyebrow label):
  - Red List category (segmented, 7 buttons: CR, EN, VU, NT, LC, DD, NE + "Any")
  - Endemic status (segmented: Endemic / Native / Introduced / Any)
  - Family (multi-select chips)
  - Basin (multi-select chips, alphabetical)
  - CARES status (segmented: CCR / Priority / Monitored / Any) — softened per §7
  - SHOAL priority (toggle: On / Any)
  - Captive coverage (segmented: Held / Not held / Any)
- Active filters render as a chip row at top of content area with a "Clear all" link
- Cards render in the grid (§15.2), roomy by default

16.4 Species Profile (`/registry/species/{slug}`)

- Full-bleed hero strip at top, 280px tall
  - Background: large species photo (`hero_photo_url`), or `--bg-sunken` stripe pattern if absent
  - Overlaid bottom-left: IUCN pill (large), scientific name in Spectral italic 56px, common name line, authority + year
- Body container `container-narrow` (max-width 920px), padding `40px 28px`
- **Three-up meta strip** directly below hero:
  - Col 1: Taxonomy (family, genus, authority)
  - Col 2: Distribution (basin chip, locality count, small inline map thumbnail 180×360)

    - Col 3: Ex-situ summary (3 stats: institutions / individuals / programs)
- **Sections**, each preceded by an eyebrow + H2:
    - *Overview* — description copy
    - *Distribution* — full map widget (§2) + locality list
    - *Ex-situ coverage* — full per-institution table
    - *Husbandry notes* — rich text
    - *Field programs* — list of partner projects with dates and links
    - *References* — citations list
- Sections separated by 48px vertical space and a 1px `--rule` horizontal line

16.5 Map page

- Full-viewport split: left rail 360px species list + filters, right fills with interactive map (§2)
- Top-right overlay legend: 7 IUCN category toggles (square 12px color + label)
- Hovering a species in the rail highlights its locality dot(s); clicking zooms + opens profile

16.6 Dashboard

- Container padding `40px 28px`, max width 1280px
- H1 "Conservation dashboard" + small subhead with last-updated timestamp
- 5 stacked sections per §6, each a `.card` with eyebrow + H2 + body content:
    1. Coverage gap panel (sortable table)
    2. Single-institution risk
    3. Studbook status
    4. Recently updated
    5. Red List snapshot (the histogram, demoted from hero)

16.7 Footer

- `--bg-sunken` background, top border `--rule`, 32px top / 48px bottom padding
- 4-column grid: brand + tagline / Explore links / About links / Colophon
- 12px caption type throughout

---

17. Spacing scale

Use only these step values. `mt-*` classes in the prototype CSS correspond 1:1.

```
4px · 8px · 12px · 16px · 24px · 32px · 48px · 64px · 80px · 120px
```

Section-to-section vertical rhythm: 56px (compact) or 80px (roomy). Inside cards: 12–16px between items.

---

18. Motion

- **Route transitions:** 200ms fade + 4px rise (`.route-fade` in prototype CSS). Apply to the main route container only.
- **Hover transitions:** 120ms on border + background + color. Never more than 200ms.
- **Focus ring:** instant.
- **Pulse dot** (live-sync indicator): 1.8s infinite ease-in-out opacity 0.4→1.0, no scale.
- No parallax, no scroll-driven animations, no entrance animations on page load beyond the route fade.

---

19. Accessibility additions to §10

- Color contrast: ink/bg is WCAG AAA (21:1), ink-2/bg AA large (7:1), ink-3 AA small (4.5:1 on --bg, fail on --bg-raised for body — reserve ink-3 for 12px+ captions).
- All IUCN pills must include the two-letter code as text.
- Focus ring 2px `--focus`, 2px offset.
- Never rely on color alone to indicate status.

---

20. Reference screenshots

Captures of the prototype at the locked-in default theme live in `handoff-refs/`. Use these as the visual source of truth when a spec detail is ambiguous.

- `handoff-refs/01-page.png` — Home / Overview (stat hero + three-up + sync strip)
- `handoff-refs/02-page.png` — Species Directory (filter rail + card grid)
- `handoff-refs/03-page.png` — Species Profile (Paretroplus menarambo)
- `handoff-refs/04-page.png` — Map page (split layout)
- `handoff-refs/05-page.png` — Dashboard

These are pixel-accurate to the prototype. Production should match layout, proportions, and density, but swap in the real map widget and real silhouettes per §2–§3.

Quick prompt for the coding agent

> *Port the prototype at `Madagascar Freshwater Fish redesign.html` (source in `prototype/`) into the WPS production codebase. Follow both parts of `Developer Handoff.md`.*
>
> ***Part 1 — data & plumbing:***
>
> 1. *Replace the `MadagascarOutline` component everywhere it's used with the existing WPS site map widget. Locality dots must come from the canonical locality endpoint, filtered by species ID where the prototype passes a single-species context.*
> 2. *Replace `FamilySilhouette` with a new `SpeciesSilhouette` that resolves species → genus → family silhouette URLs from the registry media library. Add `silhouette_url`, `hero_photo_url`, and credit fields to the species / genus / family CMS models.*
> 3. *Wire all other components to the Species schema in §4 — no mock data. Hydrate from a single `/api/registry/species.json` endpoint.*
> 4. *Restore common names, husbandry notes, field programs, and full ex-situ roster on the species profile.*
> 5. *Restructure the dashboard around the five ex-situ coordinator sections in §6.*
> 6. *Soften CARES and SHOAL visual weight per §7.*
> 7. *Move to real paths under `/registry/*` (§9).*
> 8. *Pass the a11y checklist in §10 + §19.*
>
> ***Part 2 — visual system:*** *9. Bake the design tokens in §12 into production CSS custom properties. Drop all other prototype themes and the Tweaks panel. 10. Load Spectral + IBM Plex Sans + IBM Plex Mono as web fonts. Apply the type roles in §13 strictly. 11. Build components per §15 (IUCN pill, species card, filter chips, segmented control, inputs, buttons, basin pill, coverage-gap stat). Match variant states exactly. 12. Compose pages per §16. Use the stat hero on the welcome page. Respect the 1280px max-width, the spacing scale in §17, and the motion rules in §18. 13. Compare every page visually against the*

> *reference screenshots in `handoff-refs/`. Flag any deviation for design review rather than resolving it unilaterally.*